Linux Administration Commands Sheet

1. File & Directory Management

- `ls -lah` : List all files (including hidden), with detailed permissions, in human-readable sizes (KB, MB).
- `pwd` : Print working directory (shows your current absolute path).
- `cp -r /source /dest` : Recursively copy a directory and its contents.
- `mv /source /dest` : Move or rename a file/directory.
- `rm -rf /dir` : Forcefully and recursively remove a directory. *(Use with extreme caution!)*
- `find /path -name "filename"` : Search for a specific file starting from a given path.
- `grep -rnw '/path' -e 'pattern'` : Search for a specific text string inside all files within a directory.
- `tar -czvf archive.tar.gz /dir` : Compress a directory into a gzip archive.
- `tar -xzvf archive.tar.gz` : Extract a gzip archive.

2. System, Memory, & Storage

- `top` or `htop` : Real-time view of running processes, CPU, and RAM usage (`htop` is much more user-friendly but may need to be installed).
- `free -m` : Show available and used memory (RAM and Swap) in Megabytes.
- `df -Th` : Display disk space usage and file system types in human-readable format.
- `du -sh /path/to/dir` : Calculate the total disk space utilized by a specific directory.
- `uname -r` : Display the current Linux kernel version.
- `dmesg -T | tail -n 20` : View the last 20 kernel ring buffer messages (excellent for troubleshooting hardware or driver failures).

3. Networking (Modern `iproute2` suite)

- `ip a` (or `ip addr`) : Show all IP addresses and network interfaces (replaces the deprecated `ifconfig`).
- `ip route` : Display the routing table (replaces `route`).
- `ping -c 4 google.com` : Send 4 ICMP echo requests to test connectivity.
- `ss -tulpn` : Show all listening ports and the processes using them (the modern replacement for `netstat`).
- `curl -I http://domain.com` : Fetch the HTTP headers of a website to verify web server response.
- `dig domain.com` : Query DNS records for a domain.

4. User & Group Management

- `id username` : Display user ID (UID) and group IDs (GID) for a specific user.
- `useradd -m -s /bin/bash newuser` : Create a new user with a home directory (`-m`) and set their default shell to bash.
- `passwd username` : Set or change a user's password.
- `usermod -aG sudo username` : Add a user to the `sudo` (or `wheel` on RedHat/CentOS) group to grant admin privileges.
- `cat /etc/passwd` : View a list of all registered users on the system.

5. Process Management

- ps aux : Display a snapshot of every process currently running on the system.
- ps aux | grep "nginx" : Filter the process list to find a specific application.
- kill <PID> : Send a graceful termination signal (SIGTERM) to a process ID.
- kill -9 <PID> : Forcefully kill a hung process (SIGKILL).

6. Permissions & Ownership

(Note: Linux permissions use Read=4, Write=2, Execute=1)

- chmod 755 filename : Set permissions so the owner can read/write/execute (7), and everyone else can only read/execute (5).
- chmod 600 id_rsa : Secure an SSH key so only the owner can read/write it.
- chown user:group filename : Change the user and group ownership of a file or directory.
- chown -R user:group /dir : Recursively change ownership for a directory and everything inside it.

7. Service Management (systemd)

- systemctl status service_name : Check if a service (like nginx or sshd) is running, crashed, or stopped.
- systemctl restart service_name : Restart a service to apply configuration changes.
- systemctl enable service_name : Configure a service to start automatically when the server boots up.
- journalctl -u service_name -f : Follow the live logs for a specific systemd service (crucial for debugging why a service won't start).

Verification & Cross-Checking Note:

As a best practice, if you are ever unsure what a command does or what flags are available, you don't need to guess. You can always cross-check locally on your terminal using:

- man <command> (e.g., man chmod) to open the full manual.
- <command> --help (e.g., mkdir --help) for a quick summary of arguments.